

A second observation worth measuring

SCILLA PASSIVE asks a narrow industrial question: can an existing maritime track use one uncertainty-bearing observation derived from non-cooperative merchant-radar illumination?

The corrected process-model audit preserves the architecture result across all 81 frozen stress gates. The next claim cannot come from another simulation. It must come from a partner-controlled capture or replay.

81**stress gates rerun**

No architecture-gate reversal after process correction.

94.4%**physics cells favourable**

51 of 54 simulated cells beat propagation only.

49.0%**EIG beats shortest pulse**

No robust optimizer advantage is claimed.

THE PARTNER ASK

Run one bounded falsification test.

Use partner-owned RF hardware or frozen maritime data. Preserve the existing baseline. Predeclare the gates. Stop if the observation is not repeatable, integrity-safe or economically useful.

Prepared 2 September 2026 · Michele Giletto · michele.giletto@gmail.com

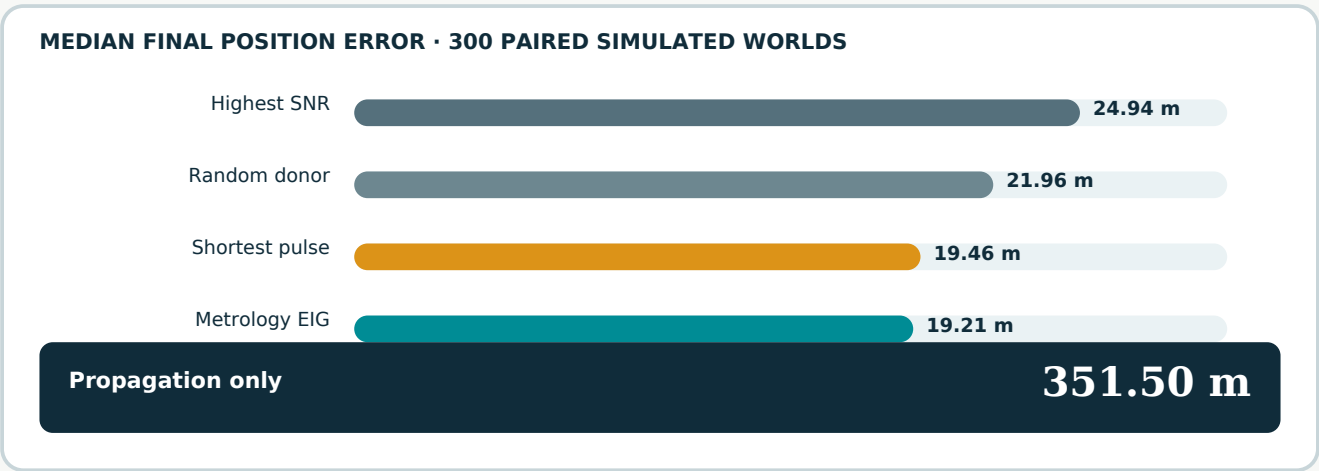
Public evidence: github.com/michelegiletto-beep/scilla-passive-evidence

Scientific record: doi.org/10.5281/zenodo.22229086

EVIDENCE AUDIT

The architecture signal survived.

The immutable 1.0.0 release contained a partition-dependent process-noise formulation. Maintained main discloses it, implements a continuous-white-noise-acceleration candidate, and reruns the complete evidence ladder.



51 of 54

physics gates favourable

Zero gate changes versus the legacy model.

27 of 27

integrity gates favourable

Zero gate changes versus the legacy model.

0.0 m

paired median EIG difference

The interval spans zero; optimizer moat not supported.

Decision boundary

SUPPORTED IN DECLARED SIMULATION

A valid second observation can sharply reduce uncertainty in the declared cue-driven fixtures.

NOT YET DEMONSTRATED

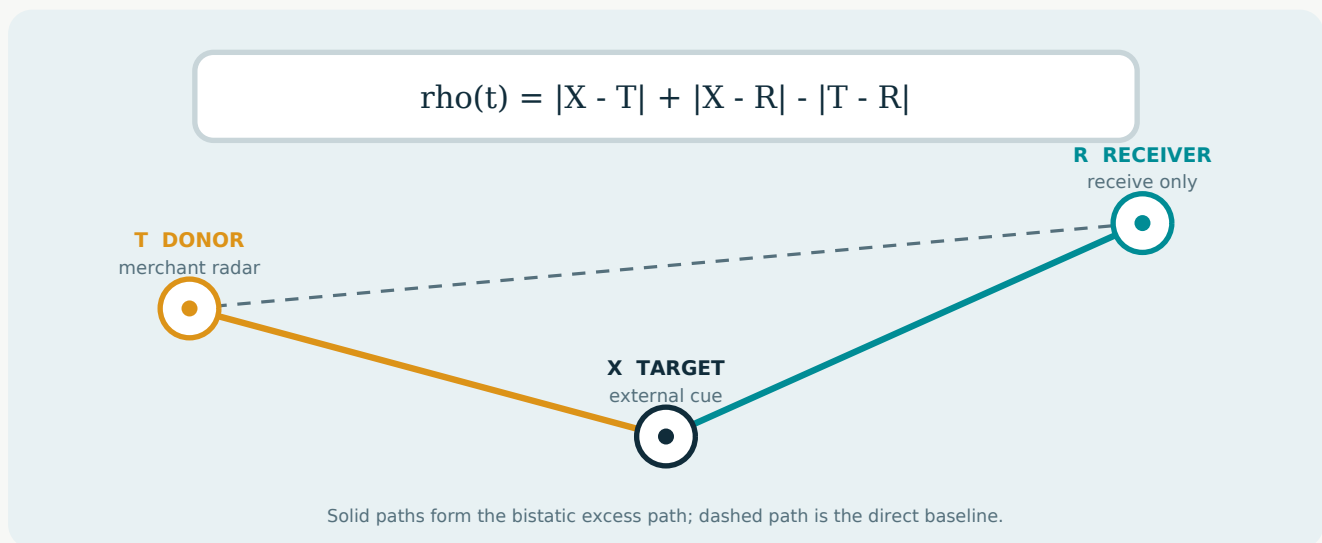
RF feasibility, sea clutter, association integrity, operational value and integration economics.

Candidate status: PASS_CANDIDATE_NOT_PROMOTED. Numbers on this page are simulation outputs, not measured range or detection performance.

MEASUREMENT ARCHITECTURE

One cue. One excess path. One decision.

The receiver does not control the donor radar and does not perform blind search. It asks whether a donor, target and receiver geometry supports one bounded second observation.



Minimum useful evidence

- 01 Same known target**
The cue exists before passive admission.
- 02 At least 10 independent sweeps**
Rejected sweeps remain visible.
- 03 Geometry-compatible delay**
Uncertainty includes donor metrology.
- 04 Incremental track value**
Compared with a frozen operational baseline.

A positive result is not “we detected a ship.” It is a repeatable, uncertainty-bearing observation that changes a predeclared tracking decision.

VALIDATION DESIGN

Two ways to answer the next question.

A partner may contribute coherent RF access or a frozen maritime replay. Neither path requires the partner to buy a new surveillance programme.

TRACK A

Coherent capture

Partner-owned dual-channel receive chain, known donor and truth-referenced target.

- Reference and surveillance channels
- Timing and cable-delay calibration
- Ten-sweep excess-path test

TRACK B

Frozen maritime replay

Partner-controlled cues, donor metadata and retained measurements or signal data.

- Metrics frozen before replay
- Existing baseline preserved
- Truth withheld from the algorithm

Acceptance ladder

0

Calibration

Coherence, timing, saturation and receive-only state.

1

Association

Donor identity and propagated position uncertainty.

2

Repeatability

Ten independent sweeps without post-hoc thresholds.

3

Track value

Improvement against the partner's frozen baseline.

4

Economics

Integration cost remains below plausible operational value.

Final outcomes are PASS, REDESIGN, KILL or INSUFFICIENT EVIDENCE. A failed gate is a useful result when it prevents unbounded integration spend.

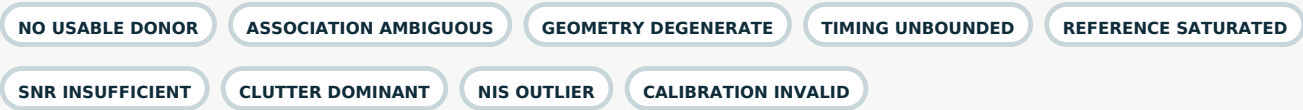
SYSTEM BOUNDARY

A measurement service, not a replacement radar.

SCILLA PASSIVE receives an external cue and one candidate illumination opportunity. It returns one bounded observation or an explicit rejection reason.



Required rejection vocabulary



Insertion points

SYSTEM	ROLE	NON-NEGOTIABLE BOUNDARY
Track manager	Supplies cue and covariance	No silent state overwrite
Signal processor	Produces delay and uncertainty	Calibration retained
Fusion engine	Admits bounded observation	Baseline replay preserved
Operator view	Shows evidence and reasons	Simulation and measurement distinct
Evidence store	Preserves inputs and hashes	Deterministic audit trail

Every output carries its software commit, model version, calibration ID and input hash. Missing covariance, timing or association must fail safe.

PARTNER DECISION

The acquisition thesis must be earned.

A public simulation package is not a product and not an exit. It can, however, make a low-cost diligence decision possible for an organisation that already controls the missing hardware, data or integration context.

A useful partner contributes one missing asset

RADAR OEM

Reference channel and emitter expertise

SURVEILLANCE INTEGRATOR

Frozen tracks, truth and baseline workflow

MARITIME OPERATOR

Operational decision and economics

RESEARCH LAB

Coherent SDR capture and independent reproduction

Before licensing or acquisition, answer six questions

- 01** Which existing decision becomes materially better?
- 02** What active dwell, revisit or operator action can be deferred?
- 03** What does one accepted observation cost, including rejects?
- 04** What false-association cost is operationally tolerable?
- 05** Can integration avoid a safety-critical control-loop change?
- 06** Is licensing the evidence and know-how cheaper than rebuilding it?

PROPOSED FIRST CONTACT

A 30-minute technical screen, then a bounded data gate.

Michele Giletto · michele.giletto@gmail.com

Public package: github.com/michelegiletto-beep/scilla-passive-evidence

Canonical archive: doi.org/10.5281/zenodo.22229086

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